

# SAMEER VENKATESH MOKHASI

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## EDUCATION

### BMS College of Engineering

B.E. Computer Science with Business Systems

- CGPA: 8.0

Bangalore, India

2024 -- 2027

### KLE C. I. Munavalli

Diploma in Computer Science

- Percentage: 94%

Hubballi, India

2021 -- 2024

### Chinmaya English Primary School

Class X

- Percentage: 78%

2021

## EXPERIENCE

### Athreya Technologies Private Limited

Jan 2024 - May 2024

Network Security Engineer Intern -- Spoof Detection & Network Infrastructure

- Engineered a full-stack spoof detection application using Java Swing and web frontend technologies, reducing false-positive network alerts by centralizing packet visualization across all monitored nodes.
- Audited and validated 100% of network packet transmissions across communication endpoints using real-time monitoring tools, achieving zero data-integrity failures throughout the 5-month engagement.
- **Key Skills:** Git, Java Swing, SQL, Cisco Packet Tracer, Network Security, TCP/IP

## PROJECTS

### Adaptive Traffic Signal Optimizer

Python, FastAPI, LangGraph, Claude API, React.js, Redis, PostgreSQL, WebSocket, Docker Compose

Apr 2026

- Architected a 4-agent AI pipeline using LangGraph and Claude API to dynamically optimize traffic signal timings, achieving a 28% average reduction in intersection wait times vs fixed-cycle baselines.
- Designed a stateful LangGraph pipeline (Ingestion → Congestion Analysis → Signal Optimizer → Reporter) processing live vehicle counts every 5s; persisted state to Redis/PostgreSQL, streamed results to React + Recharts dashboard via WebSocket.

### ForgeCI: Distributed CI/CD Engine

Python, React.js (TypeScript), Prisma, PostgreSQL, Docker, JWT, GitHub Webhooks, Canvas API

Jan 2026 – May 2026

- Developed a distributed CI/CD engine with a custom master-worker architecture that ingests live GitHub webhooks and orchestrates automated build pipelines across multiple execution nodes.
- Engineered a priority-based job scheduler with anti-starvation mechanisms; visualized live queue state through an interactive React dashboard with custom animations.

### VOYAGO

Node.js, Express, React.js (TypeScript), Prisma, Docker, PostgreSQL, GitHub Webhooks, Canvas API

Aug 2025 - Dec 2025

- Architected a scalable ride-hailing backend with FastAPI and PostgreSQL, delivering on-demand and pre-scheduled booking flows through a modular API design supporting concurrent request handling.
- Implemented RESTful endpoints with Pydantic validation; integrated WebSocket connections for real-time driver-rider state synchronization, reducing latency in ride acceptance events

## TECHNICAL SKILLS

**Languages:** C/C++ -- DSA (Proficient), Python, SQL

**Web & Backend:** React.js, FastAPI, HTML5, CSS3, Full Stack Development, Docker, WebSocket, Redis, CI/CD

**AI & Machine Learning:** LangGraph, Multi-Agent Systems, Claude API, YOLOv8, OpenCV, Isolation Forest, LightGBM, LSTM, SHAP (Explainability)

**Soft Skills:** Cross-functional Collaboration, Technical Communication, Product Thinking, Stakeholder Presentation

## ACHIEVEMENTS

- Generative AI: Completed 21 specialized GenAI courses covering LLMs, prompt engineering, RAG, and fine-tuning.
- Competitive Programming: Earned CodeTantra badge for completing 10 sessions in competitive programming.

## CERTIFICATIONS

- Internship Certificate in Cybersecurity -- Athreya Technologies Private Limited
- Nutanix cloud certification -- Nutanix
- AI Certification -- IBM